Powertrain comparison: run and change data

One model file | Fuel-cell electric drive and conventional diesel reference

Use **Powertrain_Comparison.slx**. It contains the driving inputs, both powertrains, battery management, energy accounting, editable data and result scopes. Subsystems are named by their roles.

Run it in five steps

1. Close the older model and open **Powertrain_Comparison.slx** in MATLAB R2024b with Simulink.

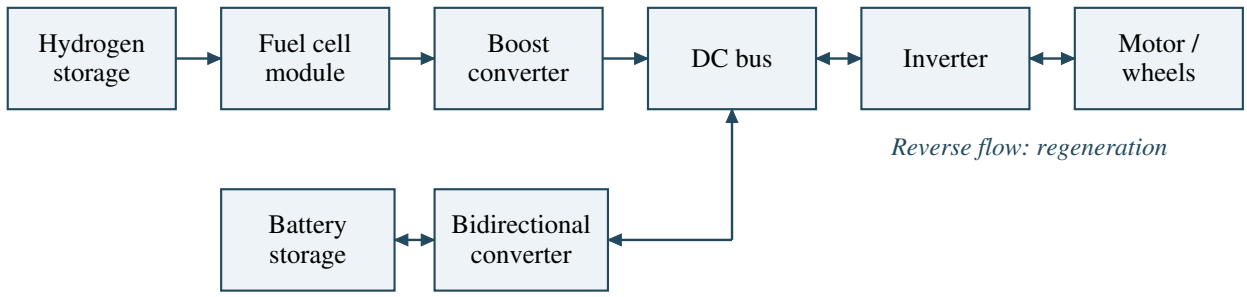
2. Open **Operating and component data**. Double-click a blue Constant block and change its **Value**. No setup script or separate data file is needed.

3. For battery settings, open **Electric powertrain > Battery storage > Battery management**. Stop the simulation before changing ratings, cell configuration or initial conditions.

4. Set **Stop time** on the toolbar and click **Run**. The default is **3600 s**. Keep the fixed solver step at **1 s**. Save the model to retain edited data.

5. Open **Results dashboard**. Read the values and double-click a Scope for its time history. Scope signal legends identify each trace. Check the feasibility and cost-valid flags before interpreting savings.

How the concept is represented



Physical energy flow: fuel cell supports the shared DC supply.

The fuel cell converts hydrogen energy into electricity. The battery supplies the remaining demand, accepts spare fuel-cell power and stores recovered braking energy. Both feed the motor through the DC bus and inverter.

The diagram shows physical energy flow. Simulink signal lines carry demand, feedback and state values. The diesel branch calculates engine power and fuel use for the same speed, gradient and passenger load, using its own vehicle mass.

This replacement contains its calculations and data internally; it needs no external setup/check functions. Repaired and tested locally in MATLAB R2024b. See page 7 for the verification record.
Verified 25 October 2026. No external setup script or startup rewiring is required.

Operating and vehicle data

Open Operating and component data. Defaults are study settings, not a measured route.

Operating input	Default	Meaning
Target speed	80 km/h	Entry range 0-100 km/h.
Speed profile	1	1 = built-in cycle; 0 = constant target.
Road gradient	0%	Positive uphill; input limited to -20% to +20%.
Auxiliary electrical load	3 kW	Input limited to 0-20 kW.
Ambient temperature	30 deg C	Input to the battery thermal model.
Fuel cell enable	1	1 = automatic range extension; 0 = battery-only operation.
Requested acceleration / deceleration	1 / 1 m/s^2	Limits how quickly the commanded speed changes.

The repeated 600 s cycle ramps to the target during 0-80 s, cruises during 80-500 s, slows during 500-580 s and stops during 580-600 s. It is a demonstration cycle, **not measured Accra-Tamale driving data**.

Vehicle parameter	Default
Common passengers and luggage kg	540
Diesel kerb mass kg	2650
Converted kerb mass kg	3004
Air density kg per m3	1.18
Drag area CdA m2	1.8
Rolling resistance coefficient	0.012
Loaded wheel radius m	0.35
Electric reduction ratio	10
Electric gearbox efficiency	0.97
Gross mass screen kg	3710

Mass and comparison boundary

The diesel kerb assumption is 2650 kg, within the 2615-2680 kg range recorded for the reference vehicle in the study. Adding the common 540 kg payload gives **3190 kg**. The converted kerb estimate includes a 50 kg support allowance; 3004 + 540 gives **3544 kg**.

Both vehicles use constant mass throughout the trip. Changing battery size, motor rating or hydrogen capacity does not update the CAD or kerb mass automatically. Update the mass estimate when equipment changes. The 3710 kg screen does not verify axle loads or structural compliance.

Battery data reused from the project

Battery management retains its SOC, OCV, current-limiting, electrical-loss and thermal calculations.

Battery mask entry	Default
Series cells / parallel cells	96 / 41
Cell capacity / nominal voltage	4.2 Ah / 3.6 V
Cell resistance	0.016 ohm
Initial SOC / initial temperature	90% / 25 deg C
Cell voltage limits	[2.5 4.2] V
Discharge / charge current ratings	[2 2] C
SOC minimum, discharge reserve, charge taper, maximum	[15 20 80 95] %
Charge temperature thresholds	[0 10 35 45] deg C
Discharge temperature thresholds	[-20 0 45 55] deg C
Effective thermal mass	341.667 kg
Specific heat	900 J/(kg K)
Thermal conductance	35 W/K
Discharge / charge coulombic efficiencies	[1 1]

OCV curve: keep 13 matching entries

Curve	Values
SOC (%)	0, 5, 10, 20, 30, 40, 50, 60, 70, 80, 90, 95, 100
Cell OCV (V)	2.5, 3.1, 3.3, 3.45, 3.53, 3.6, 3.67, 3.73, 3.8, 3.9, 4.04, 4.12, 4.2

The adapted pack has **345.6 V nominal, 172.2 Ah and 59.51232 kWh nominal energy**. It is a 60 kWh-class study pack. The original model had 96s24p; this version changes parallel count to 41. Resistance becomes about 0.03746 ohm. The 2C current rating is 344.4 A before SOC, temperature and voltage restrictions.

Positive battery current means discharge; negative means charge. Actual current is limited by the battery model. Temperature rises from I^2R heating and falls through heat transfer to ambient air. The initial temperature is separate from ambient temperature.

Cell curves and thermal limits are illustrative source-model data, not measured calibration for an installed 60 kWh pack. The scaled thermal mass is an effective model value; it is not a replacement for the complete pack mass in the vehicle calculation.

Electric-drive and fuel-cell data

Open Operating and component data. The battery and fuel cell share the regulated DC link.

Electric-drive parameter	Default
Motor peak shaft power kW	120
Motor peak shaft torque Nm	300
Motor pole pairs	4
Motor PM flux Wb	0.08
Scaled phase resistance ohm	0.0064
Scaled dq inductance H	6.4e-05
Motor viscous damping Nm s per rad	0.001
Motor dq peak current A	625
Maximum modulation index	0.95
Regulated DC link voltage V	600
Inverter efficiency	0.97
Bidirectional battery converter efficiency	0.97
Regenerative wheel power limit kW	60

The saved motor project was a smaller PMSM bench model. This route model reuses its torque, dq-voltage and copper-loss equations with a steady-state assumption, $id = 0$. A current-area scale of 7.8125 changes 80 A to 625 A and divides resistance and inductance by the same factor. The 120 kW rating, conversion efficiencies and regenerative extension are study assumptions, not a calibrated commercial drive.

Fuel-cell parameter	Default
Fuel cell module rated output kW	45
Fuel cell net electrical efficiency LHV	0.5
Fuel cell boost converter efficiency	0.95
Fuel cell module ramp kW per s	2
Fuel cell ON threshold percent	60
Fuel cell OFF threshold percent	80
Fuel cell charging allowance kW	10
Initial usable hydrogen kg	2
Hydrogen LHV kWh per kg	33.3
Fuel cell nominal output voltage V	300

A 45 kW module supplies at most **42.75 kW** after the assumed 95% converter efficiency. Fuel-cell control turns on at or below 60% SOC and turns off at or above 80%. It supports the DC demand and may charge the battery. Protection can reduce power faster than the normal ramp.

Usable hydrogen is fuel mass, not tank volume or pressure. Hydrogen chemical energy becomes module electricity plus losses; the converter adds its own loss. The voltage/duty calculation uses the boost relation $V_{out} = V_{in} / (1-D)$. Switching transients are excluded.

Diesel consumption and energy prices

The conventional branch calculates fuel flow from power demand; measured L/100 km is used to calibrate it.

Diesel parameter	Default
Diesel engine rated shaft power kW	120
Diesel average brake efficiency	0.33
Diesel transmission efficiency	0.9
Alternator efficiency	0.65
Diesel idle fuel rate L per h	0.8
Diesel LHV kWh per litre	9.94769
Overrun fuel cut enable 1 or 0	1

Basic diesel calculation

$$P_{shaft,req} = \max(P_{wheel}, 0) / eta_{trans} + P_{aux} / eta_{alt}$$

$$P_{shaft} = \min(P_{shaft,req}, P_{rated})$$

$$q_{diesel} = \max(q_{idle}, P_{shaft} / (eta_{engine} \times LHV_{diesel}))$$

P is power in kW; eta denotes efficiency; q is fuel flow in L/h; LHV is fuel energy in kWh/L. Integrating q / 3600 each second gives litres used. L/100 km = 100 x litres used / commanded distance in km.

The 0.33 engine efficiency, 0.90 transmission efficiency, 0.65 alternator efficiency and 0.8 L/h idle rate are editable assumptions. Calibrate them against fuel measurements from the chosen 2019 vehicle. A full calibrated model would use engine speed/torque maps [2].

With overrun fuel cut enabled, fuel is cut during moving deceleration only when available braking power can also cover accessory demand. This is a simplified rule, not a reproduction of the vehicle ECU.

Energy-price input	Default
Electricity GHS per kWh	0
Delivered hydrogen GHS per kg	0
Diesel GHS per litre	0
Grid to stored battery charging efficiency	0.92

Enter dated electricity and diesel prices and a stated delivered-hydrogen quotation or scenario. Zero price means **not entered**; cost outputs remain -1 until all three prices are positive. Hydrogen LHV 33.3 kWh/kg and diesel LHV about 9.9477 kWh/L follow the DOE fuel-properties comparison [1].

Read outputs and interpret savings

All scopes have named signal legends. The model evaluates energy at a requested speed.

Output group	Interpretation
Battery and hydrogen	SOC, temperature, current, voltage, hydrogen remaining and hydrogen consumed.
Electric power	Fuel-cell module power, DC-bus power, battery contribution and accepted regeneration. Battery power is positive for discharge.
Motor operation	Speed, dq current and required voltage. Power, torque, current and voltage limits constrain the available drive.
Diesel fuel	Instantaneous L/h, total litres and calculated L/100 km for the same requested cycle.
Unserved demand	Electric DC, motor and diesel shortfall. A positive value means that the requested operation is not fully supported.
Comparison flags	Both drives feasible = 1 permits an energy-saving comparison. Cost comparison valid = 1 also requires entered prices and no final battery-energy surplus.
Savings	Positive saving favours the electric concept. -1 means unavailable. A genuine negative saving is possible when its validity flag is 1.

Energy and cost accounting

Battery depletion = initial stored energy - final stored energy. The battery OCV curve is integrated to calculate inventory; negative depletion means extra energy remains in the battery.

Electric net source energy = hydrogen used x hydrogen LHV + signed battery depletion.
Diesel source energy = litres used x diesel LHV.
Energy saving (%) = 100 x (1 - electric net source energy / diesel source energy).

Electric refill cost = hydrogen used x hydrogen price + max(battery depletion, 0) / grid-charging efficiency x electricity price.
Diesel refill cost = litres used x diesel price.
Cost saving = diesel refill cost - electric refill cost.

Fuel-cell electrical energy and gross battery discharge are not added again: this would count hydrogen-derived energy twice. Net source energy excludes upstream fuel production. Refill cost excludes purchase, maintenance and infrastructure costs. Extra final battery energy receives no cost credit; cost-saving output is suppressed in that case.

Data needed for a defensible feasibility result

Use verified vehicle mass and passenger load; measured baseline diesel fuel use; component resistance, efficiency, current/voltage and thermal data; a measured route speed/gradient/time profile; and dated local energy prices. The built-in cycle does not establish Accra-Tamale journey completion.

This is a one-second system-energy model. It does not predict achieved speed, detailed fuel-cell electrochemistry, PWM switching, tank pressure or installation safety.
Local execution is verified; the component and route assumptions still require calibration.

Source notes

[1] U.S. DOE, Alternative Fuels Data Center, Fuel Properties Comparison: afdc.energy.gov/fuels/properties
[2] MathWorks, Mapped CI Engine: mathworks.com/help/autoblks/ref/mappedciengine.html
Reuse basis: saved battery-management blocks and electric-drive project equations.

Local MATLAB verification

25 October 2026 | MATLAB R2024b | Powertrain_Comparison.slx

The repaired model was opened, saved, reopened and simulated on this computer. The original model structure and study parameters are retained. No external setup or repair script is needed to run the delivered file.

Repairs applied

Rebuilt and checked all 1,727 intended connections in 27 systems; regenerated block input ports; corrected the battery OCV lookup input and logical data types; restored valid block positions; removed the failing startup rewiring callback. Disconnected input/output wires and nonfinite signals are treated as errors.

Execution checks

Test	Duration	Result
Default drive cycle	3,600 s	Passed
Battery only; fuel cell disabled	3,600 s	Passed
Fuel cell startup; initial SOC 50%	1,200 s	Passed
Empty hydrogen tank at startup	1,200 s	Passed
Constant target speed, 60 km/h	1,200 s	Passed
20% uphill; shortfall correctly flagged	600 s	Passed
95% initial SOC and downhill regeneration	600 s	Passed
Hydrogen exhaustion; fuel cell shuts down	1,200 s	Passed
Entered prices; valid cost arithmetic	1,200 s	Passed
Stationary; distance comparison unavailable	600 s	Passed

Checks cover finite outputs, battery SOC limits, hydrogen mass balance, battery terminal power, diesel and electric source-energy accounting, power limits and the expected validity flags.

Default 3,600-second run

Commanded distance: 66.667 km. Final battery SOC: 61.98%. Hydrogen consumed: 0.2517 kg. Diesel consumed: 6.909 L. Both-drives-feasible flag: 1.

Test prices are illustrative inputs used only to check arithmetic. The delivered default prices remain zero. These execution checks do not establish measured vehicle performance or validate the study assumptions.